



FACULTY OF ENGINEERING
BRANCH 2 | ROUMIEH

Full Stack Development Course

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Web Design & Version Control

Chapter 4

Full Stack Development Course

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01 Version Control

02 Web Design

Web Design & Version Control

Chapter 4

Version Control

What is Version Control?

- A system that keeps records of your changes.
- Users keep the entire code and history on their local machines.
- Allows for collaborative development.
- Allows you to know who made what changes and when.
- Allows you to revert any changes and go back to a previous state.

Version Control

What is Git?

- A tool that allows you to manage and track changes to files over time.
- When you create a repository, you will see a .git directory.
- It keeps track of a file's history – it tracks changes and who made those changes.
- Each version of a file is referred to as a **commit**.



Version Control

What is Git?

- You keep your files in a repository on your local machine.
- You synchronize your repository with a repository on a server.
- If you move from one machine to another, you can pick up the changes by synchronizing with the server.
- If your partner uploads some changes to your files, you can pick those up by synchronizing with the server.



Version Control

What is a Repository?

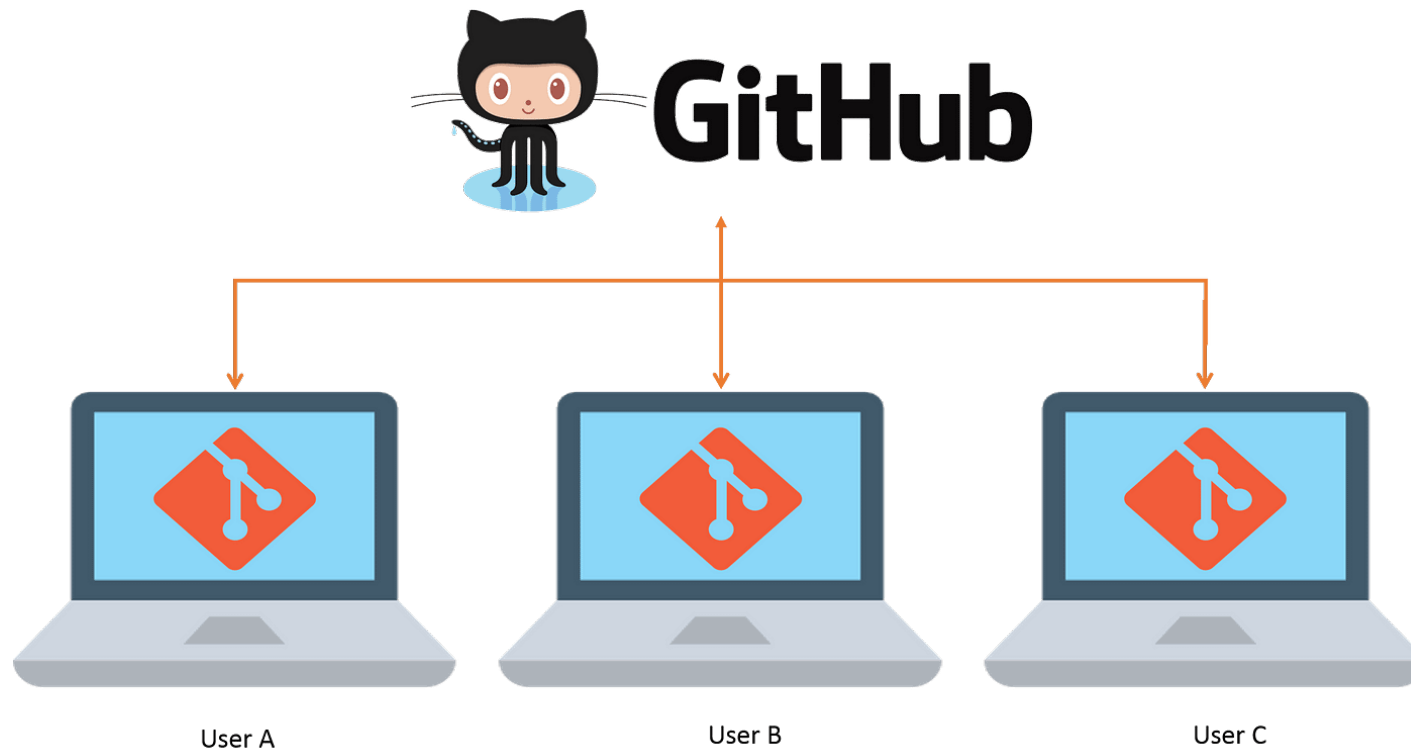
- A collection of all the files and the history of those files.
- It consists of all your commits.
- A place where all your hard work is stored.



Version Control

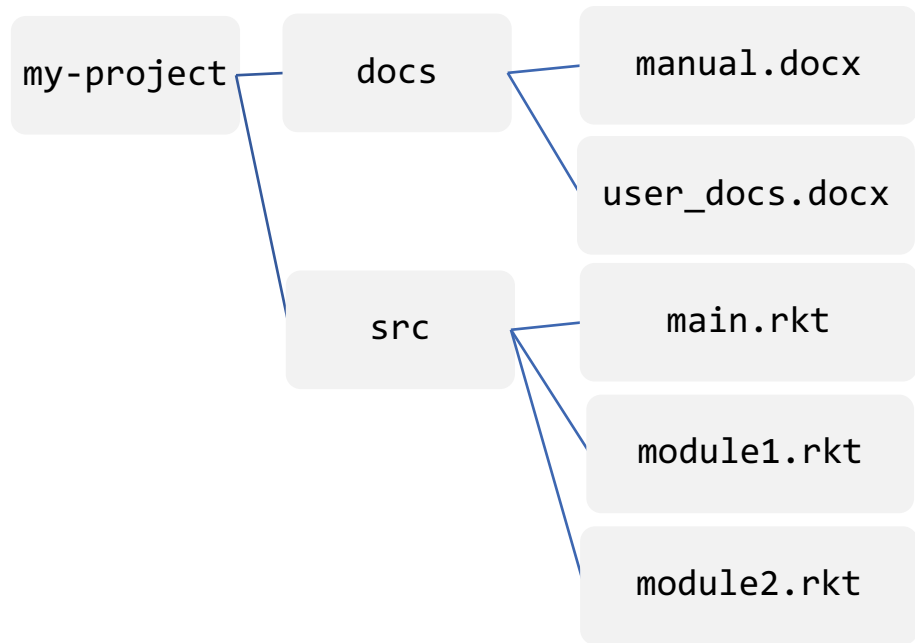
What is GitHub?

- A remote repository in which you can host your local repository.
- And a place to collaborate with others.



Version Control

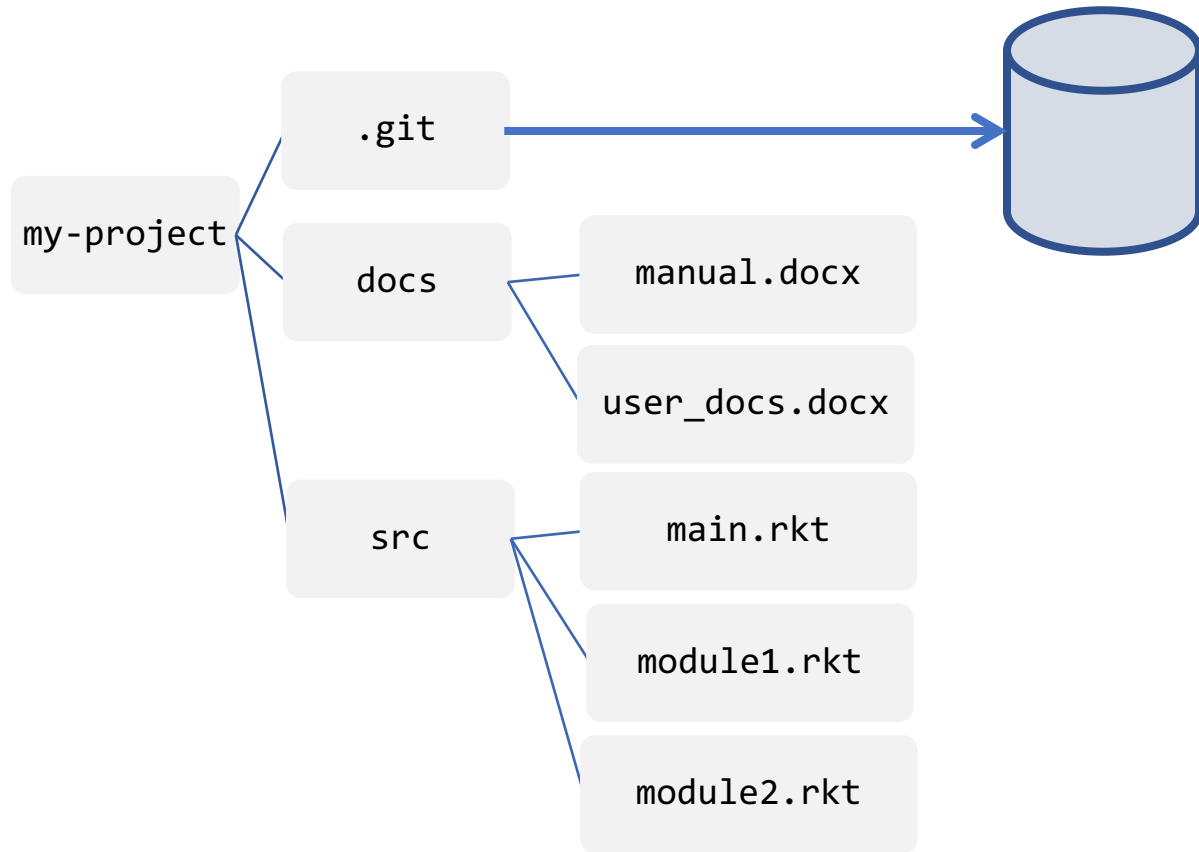
Git model



Here are your files, sitting in a directory called my-project

Version Control

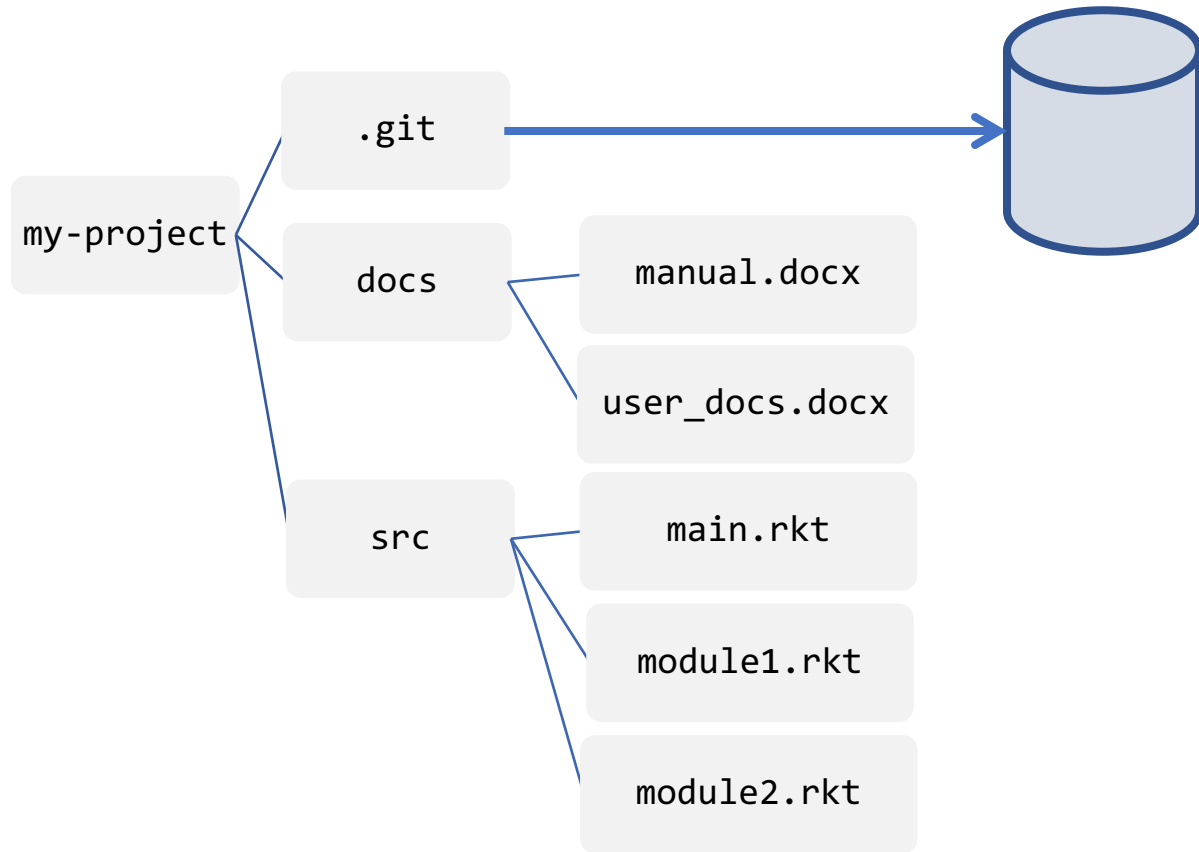
Git model



- When you have a git repository, you have an additional directory called `.git`, which points at a mini-filesystem.
- This file system keeps all your data, plus the bells and whistles that git needs to do its job.
- All this sits on your local machine.

Version Control

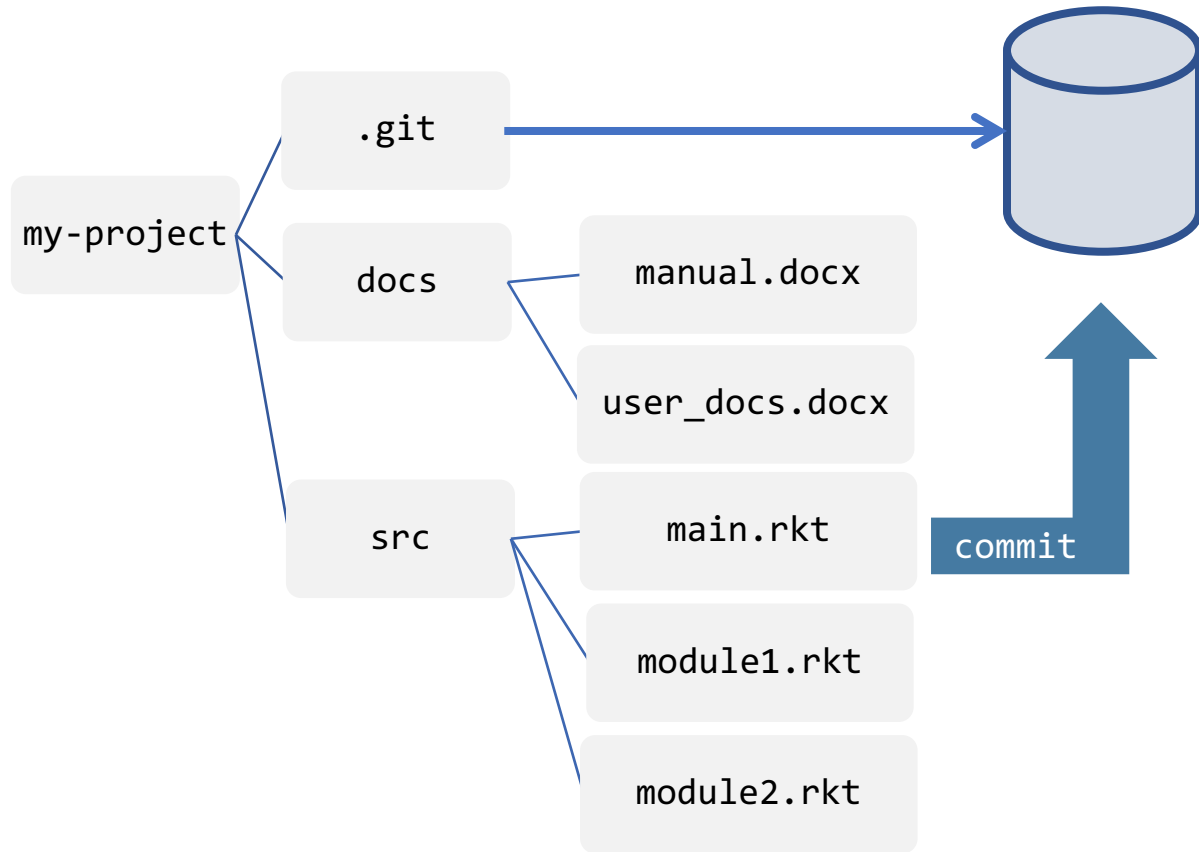
Git model



- This mini-filesystem is highly optimized and very complicated. Don't try to read it directly.
- The job of the git client (either Github for Windows, Github for Mac, or a suite of command-line utilities) is to manage this for you.

Version Control

Git model

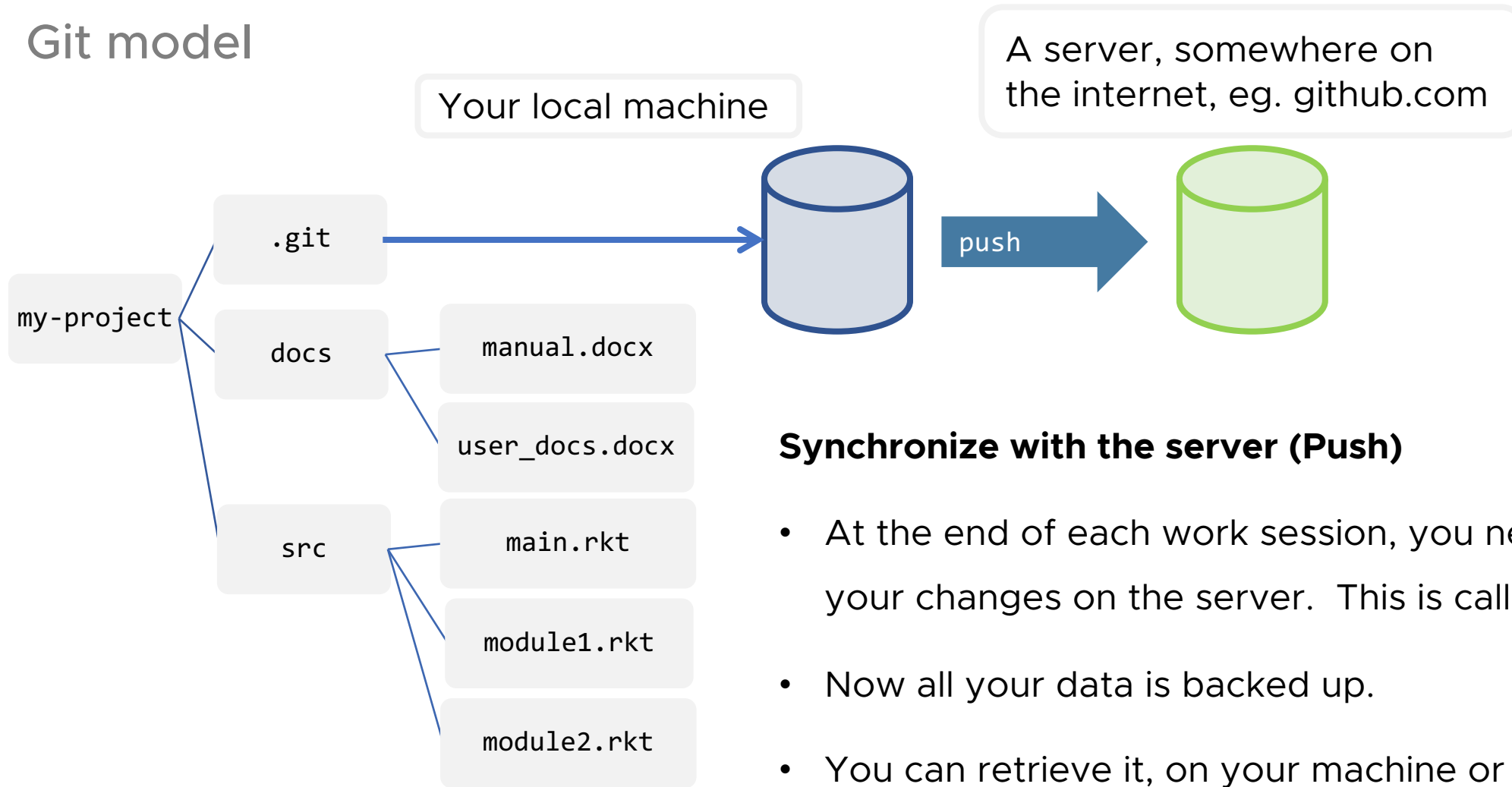


Commit

- When you do a “commit”, you record all your local changes into the mini-fs.
- The mini-fs is “append-only”. Nothing is ever over-written there, so everything you ever commit can be recovered.

Version Control

Git model

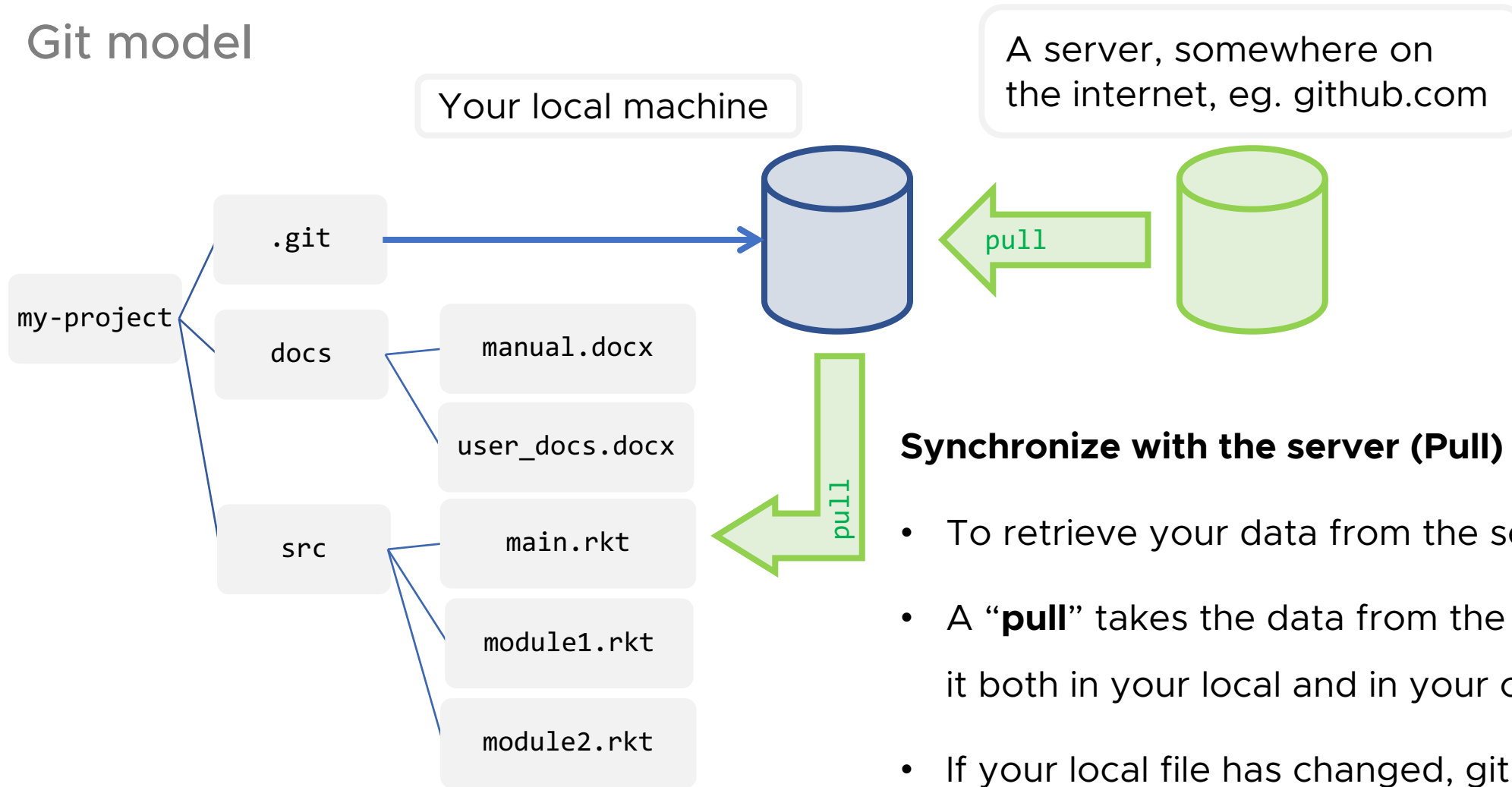


Synchronize with the server (Push)

- At the end of each work session, you need to save your changes on the server. This is called a “**push**”.
- Now all your data is backed up.
- You can retrieve it, on your machine or some other machine

Version Control

Git model

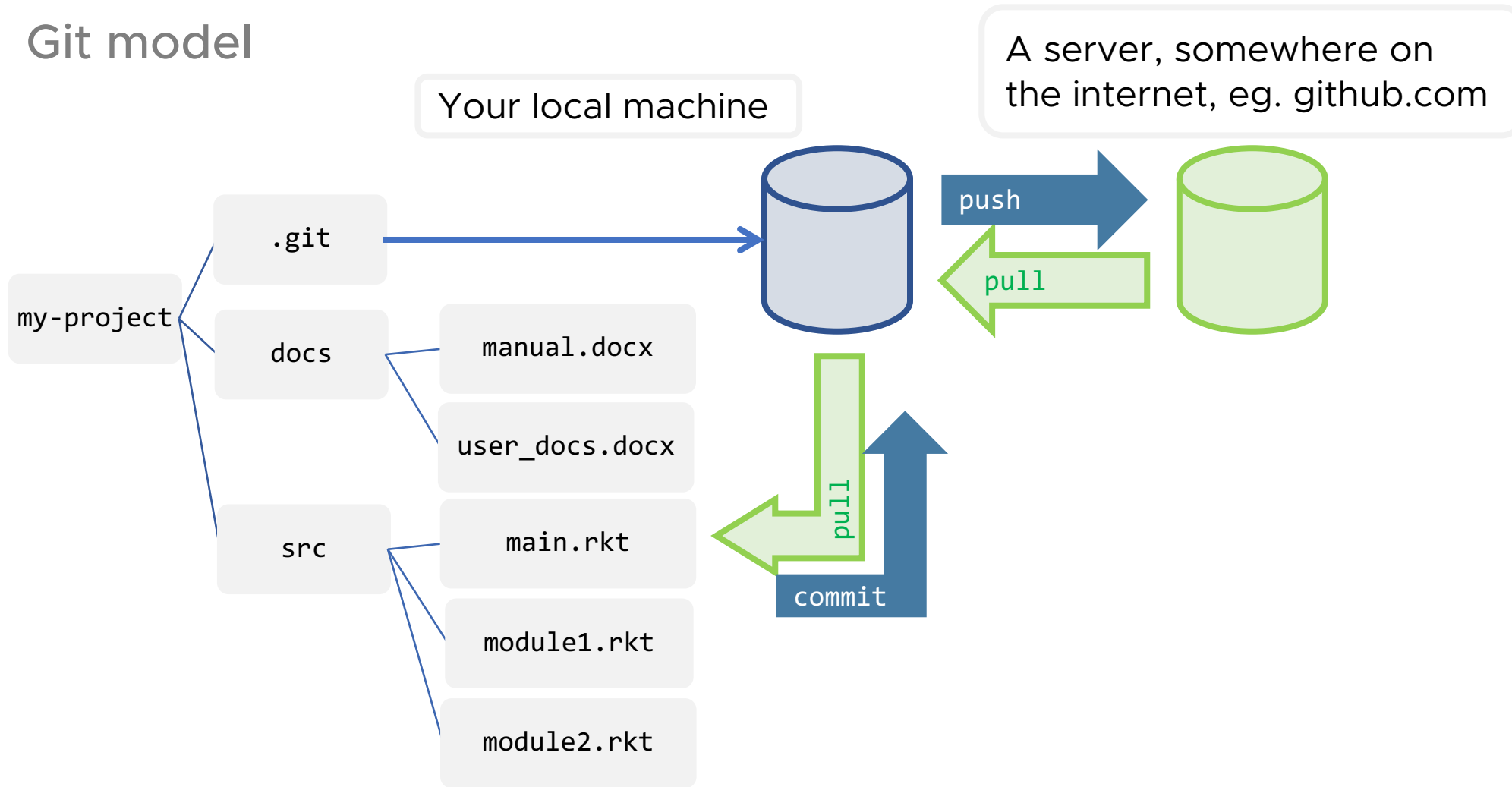


Synchronize with the server (Pull)

- To retrieve your data from the server use “**pull**”.
- A “**pull**” takes the data from the server and puts it both in your local and in your ordinary files.
- If your local file has changed, git will merge the changes if possible

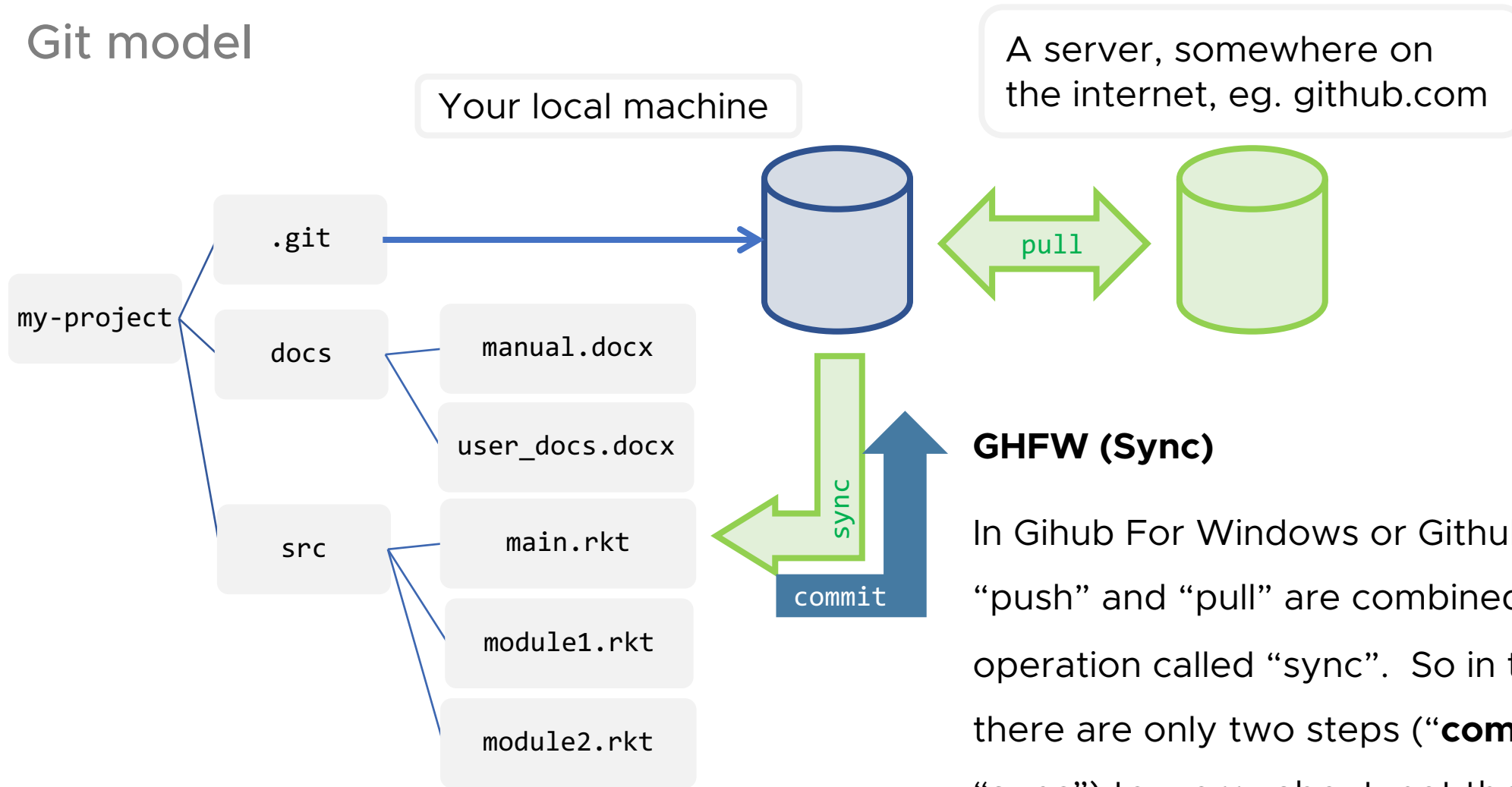
Version Control

Git model



Version Control

Git model



GHFW (Sync)

In Github For Windows or Github For Mac, “push” and “pull” are combined into a single operation called “sync”. So in these clients, there are only two steps (“**commit**” and “**sync**”) to worry about, not three.

Version Control

Git workflow

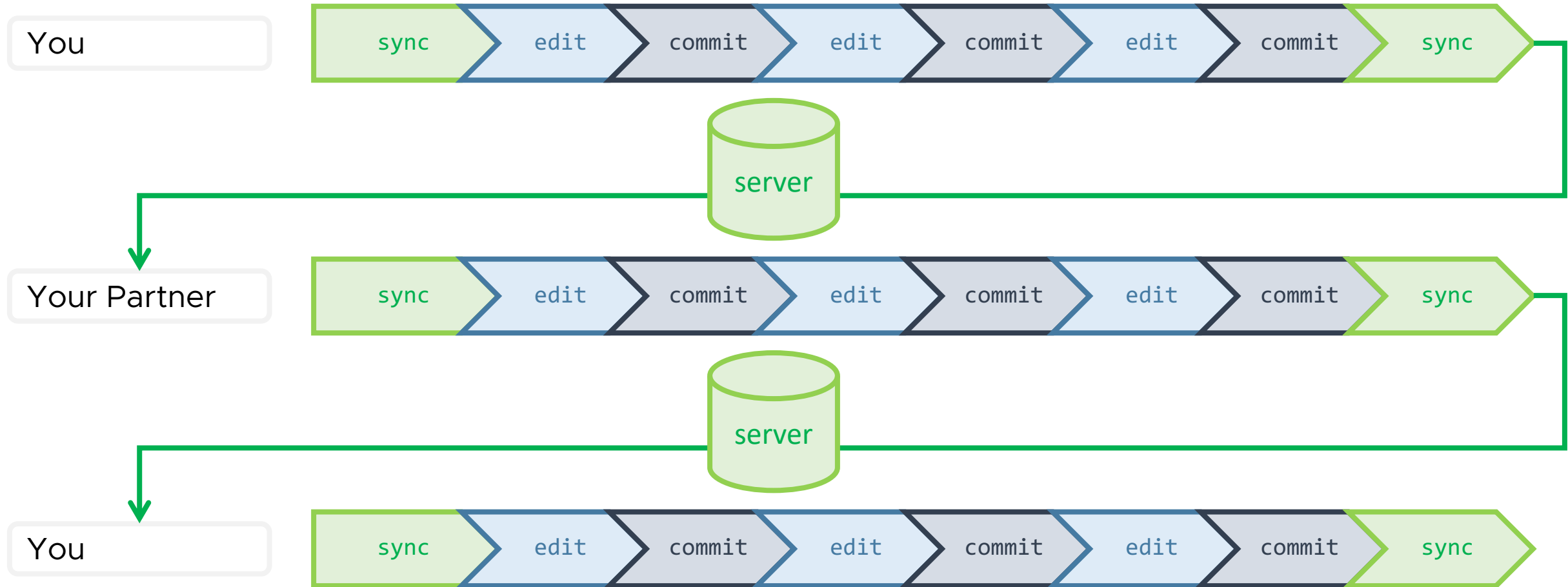
Best practice:

Commit your work whenever you've gotten one part of your problem working, or before trying something that might fail.



Version Control

Git workflow



Version Control

Branching

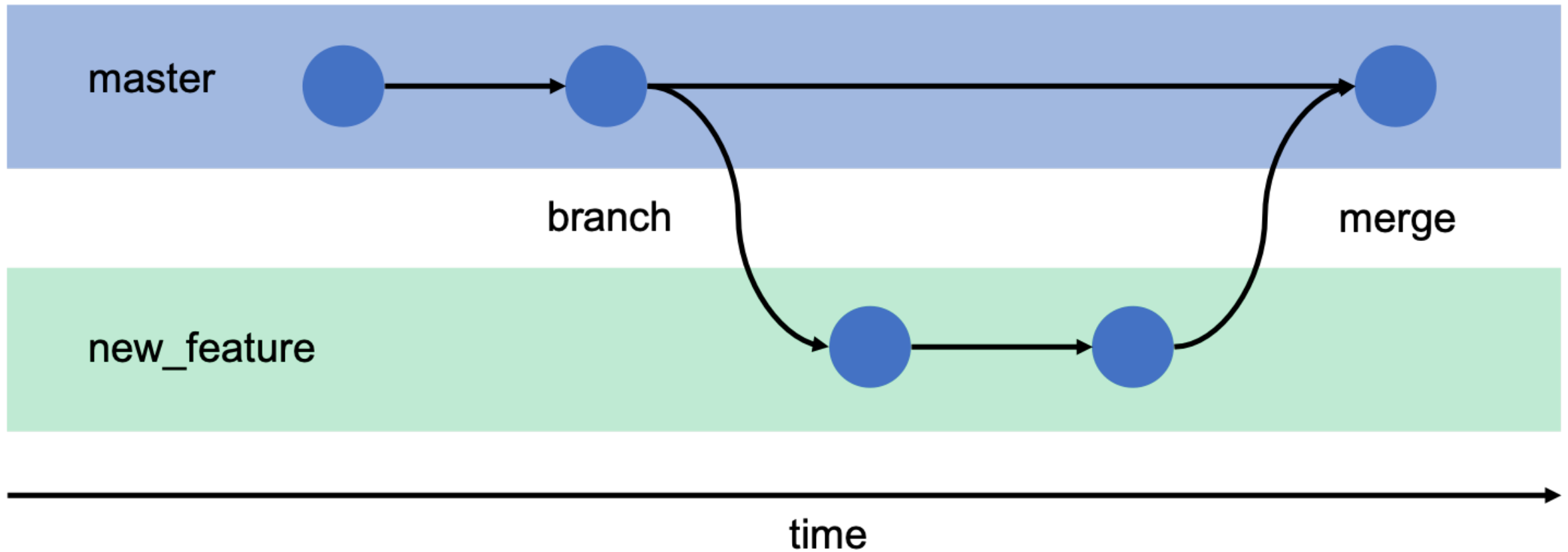
Branching allows you to create a separate line of development within your project. It's like a sandbox where you can experiment without affecting the main code.

Why Use Branches?

- Work on new features, bug fixes, or experimental ideas without touching the main branch.
- Collaborate without code conflicts.
- Keep the main branch clean and stable.

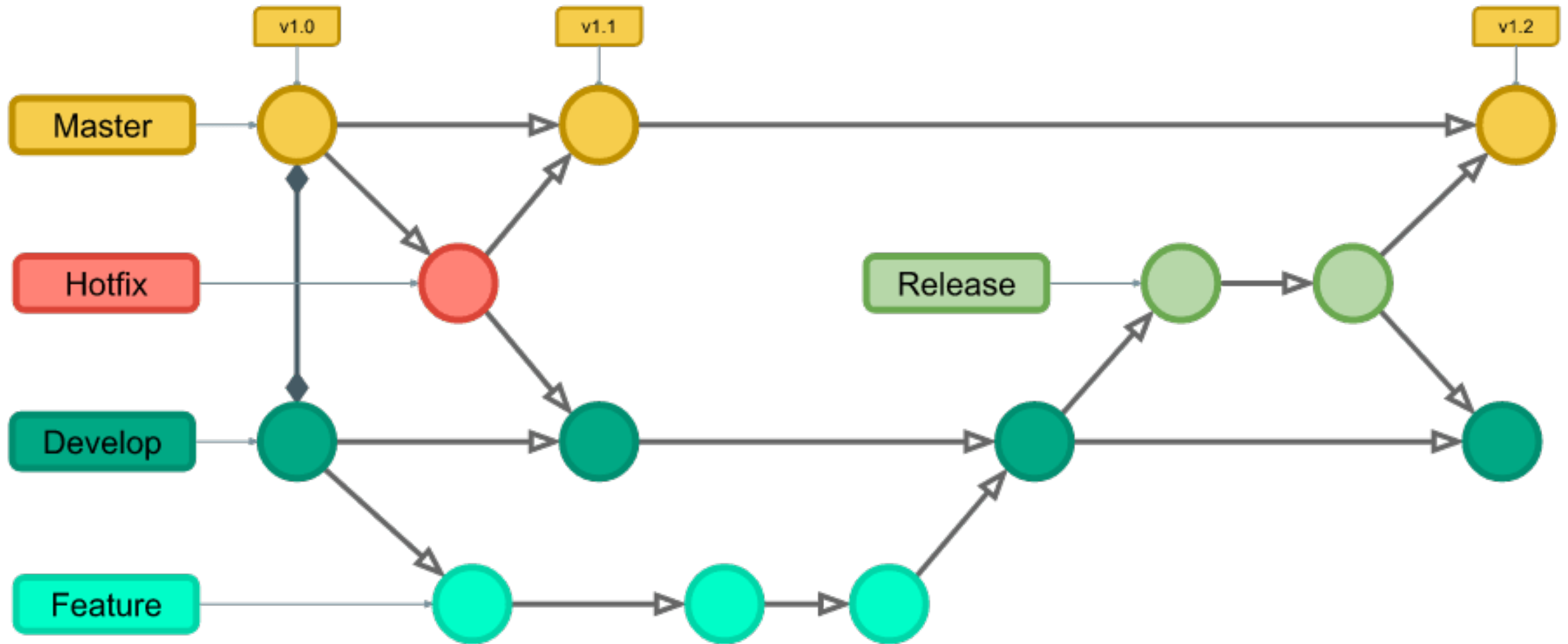
Version Control

Branching



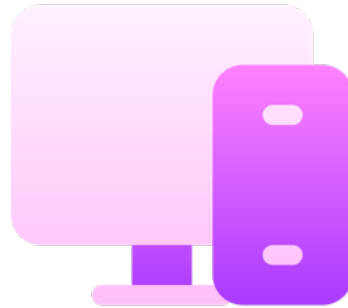
Version Control

Branching



Version Control

Demonstration



Demonstration

Git in VSCode

Version Control

What is a README File?

The README.md is the first thing people see when they visit your repository. It explains what the project is, how to use it, and why it matters.

What to Include in a README

1. **Project Title:** Clear and descriptive name of the project.
2. **Description:** What does the project do? Who is it for?
3. **Features:** List of key features or functionalities.

Version Control

What is a README File?

What to Include in a README

1. **Installation Instructions:** How to install or set up the project locally.
2. **Usage:** How to run or use the app. Include code samples or screenshots if needed.
3. **Project Structure** (optional): Explain the folder/files if it helps understanding.
4. **Contributors / Author Info:** Who built it? Add GitHub profiles or contact info.
5. **License** (optional): Add if the code is open source (e.g., MIT, GPL).

Version Control

Terminal commands

Commad	Description
<code>git clone url [dir]</code>	copy a Git repository so you can add to it
<code>git add file</code>	adds file contents to the staging area
<code>git commit</code>	records a snapshot of the staging area
<code>git status</code>	view the status of your files in the working directory and staging area
<code>git pull</code>	fetch from a remote repo and try to merge into the current branch
<code>git push</code>	push your new branches and data to a remote repository

Web Design

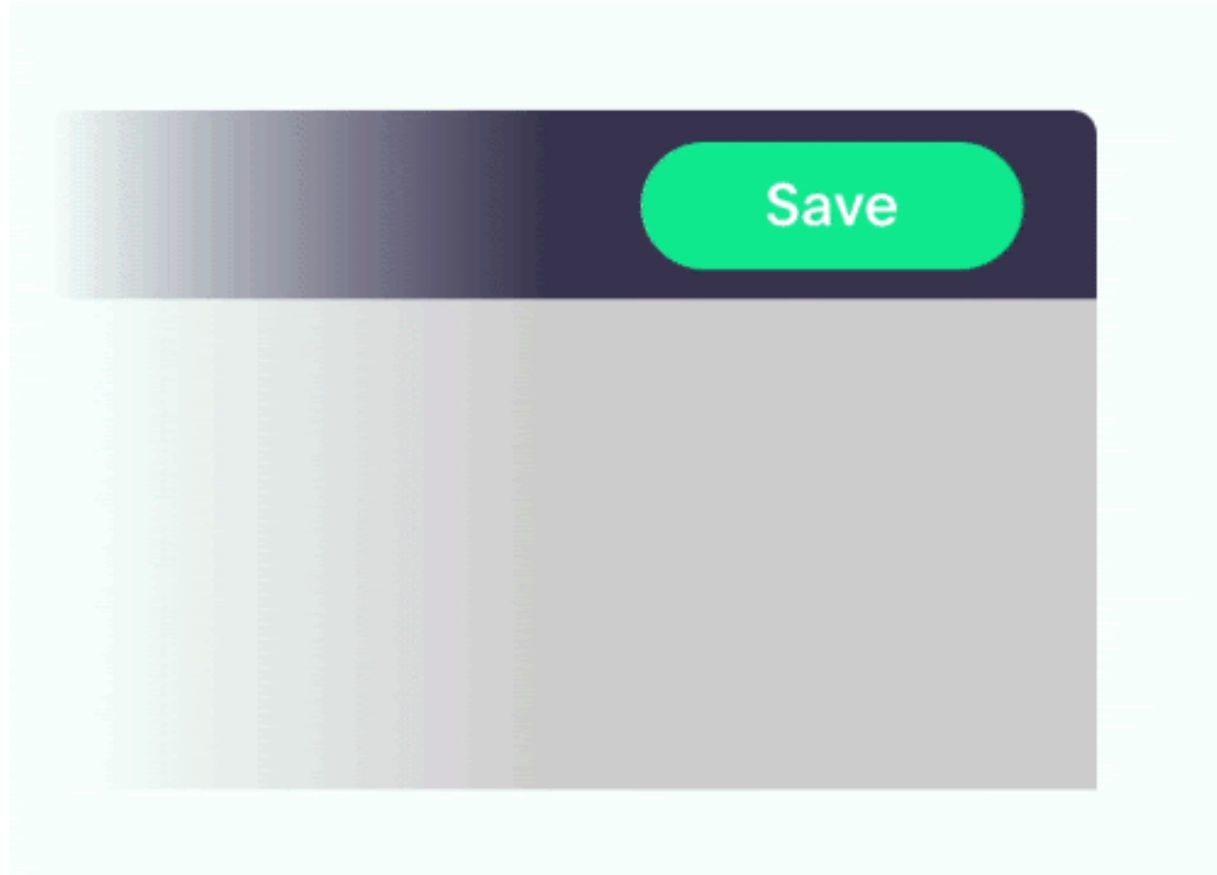
Test your design eye

QR

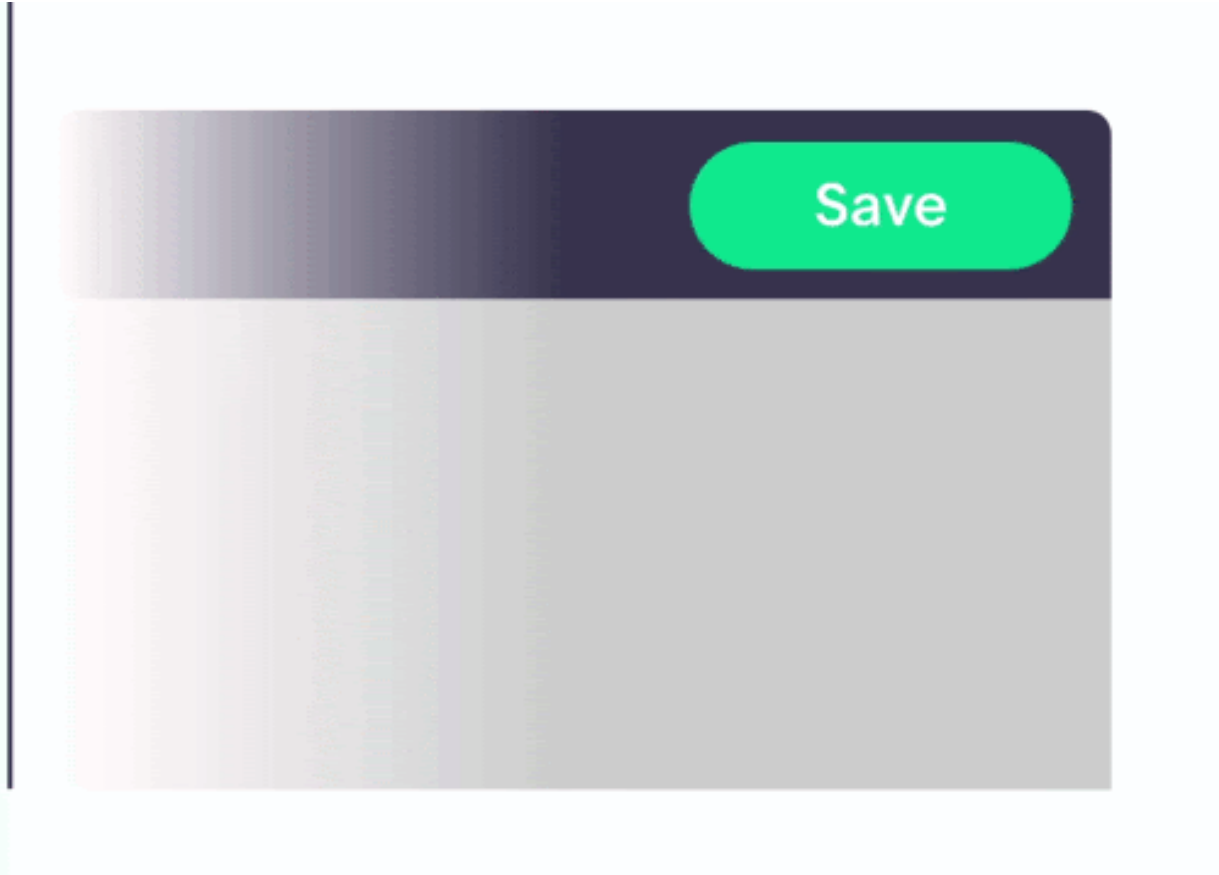


Web Design

Question 1



Option 1



Option 2

Web Design

Question 2

Initiate
autonomous
control
sequence

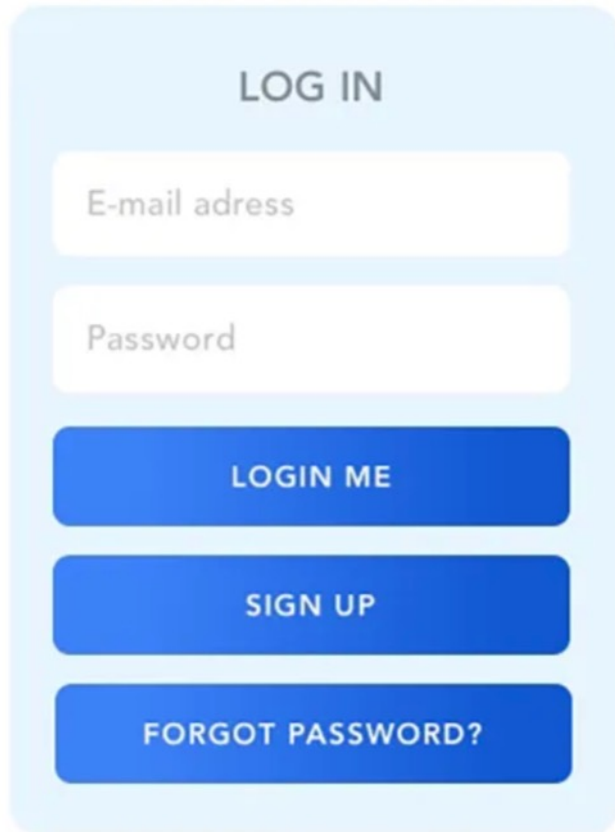
Option 1

Deploy Vehicle

Option 2

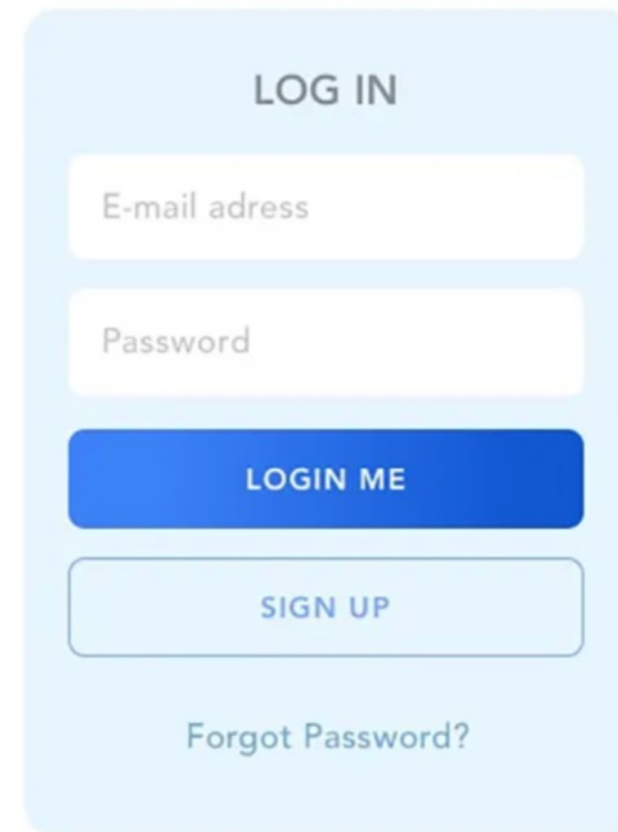
Web Design

Question 3



A login form titled "LOG IN" with a light blue background. It contains two input fields: "E-mail address" and "Password". Below the fields are three blue buttons with white text: "LOGIN ME", "SIGN UP", and "FORGOT PASSWORD?".

Option 1

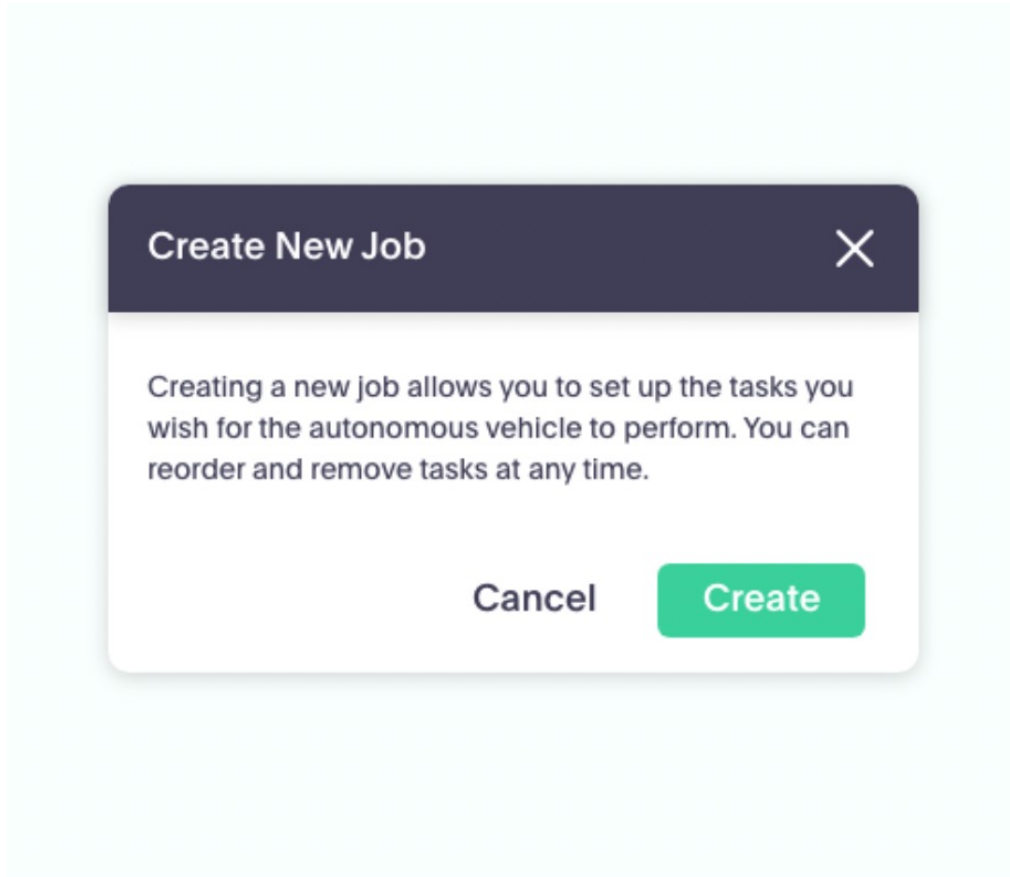


A login form titled "LOG IN" with a light blue background. It contains two input fields: "E-mail address" and "Password". Below the fields are a blue button with white text "LOGIN ME", a light blue button with blue text "SIGN UP", and a link "Forgot Password?" in blue text.

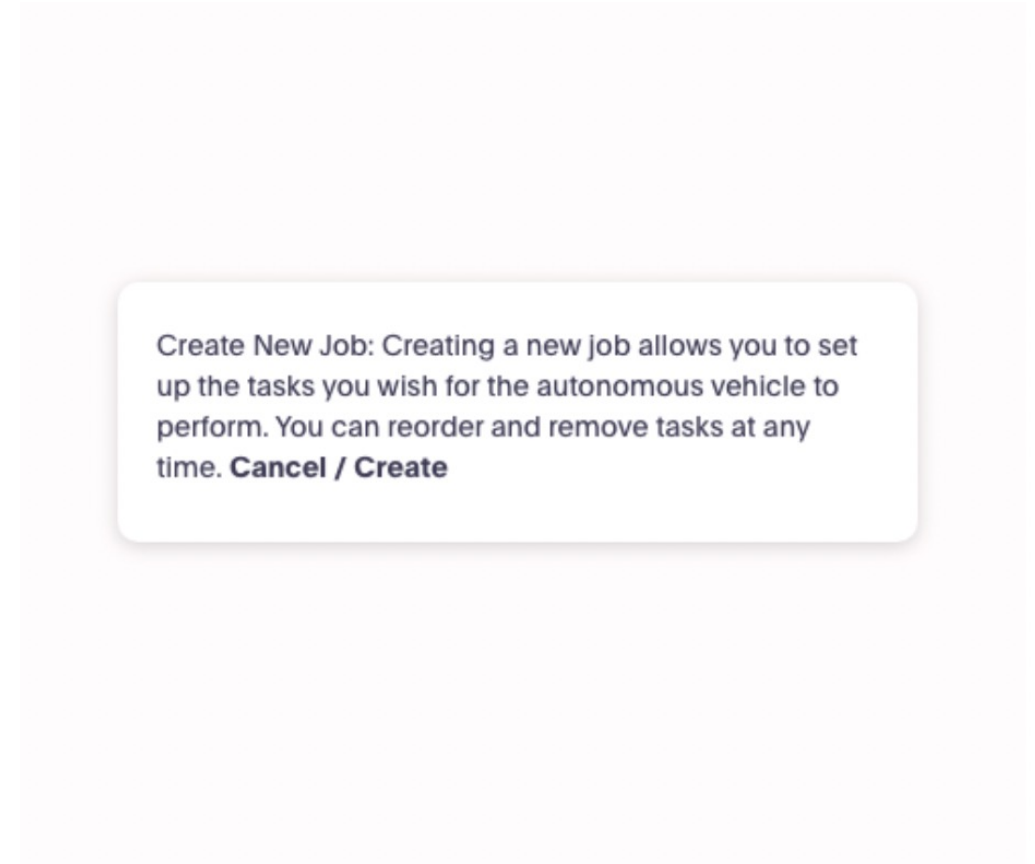
Option 2

Web Design

Question 4



Option 1



Option 2

Web Design

Question 5

First Name

First Name

Email

Oops! This doesn't look like a valid email address. Try adding '@' before the domain name. [Accepted email formats](#)

Option 1

validation fault - format does not match database.
Invalid entry.

First Name

First Name

Email

Option 2

Web Design

Question 6



Alpine Lakes High Camp ski adventures!

On an early weekday morning in late January of 2021 Jose from King 5 News program Evening Magazine, Mark Gunlogson and a group of Mountain Madness guides met up with Justin Donohue and Cedric from Alpine Lakes High Camp. We met with our ski gear and enough food for two nights and three days of backcountry skiing.

We had a short briefing about the plan for the day, and soon commenced loading up the snowcat, snowmobiles, and epic 4x4 land rover snow machines for some trail breaking.

[read more>](#)

Option 1



Alpine Lakes High Camp ski adventures!

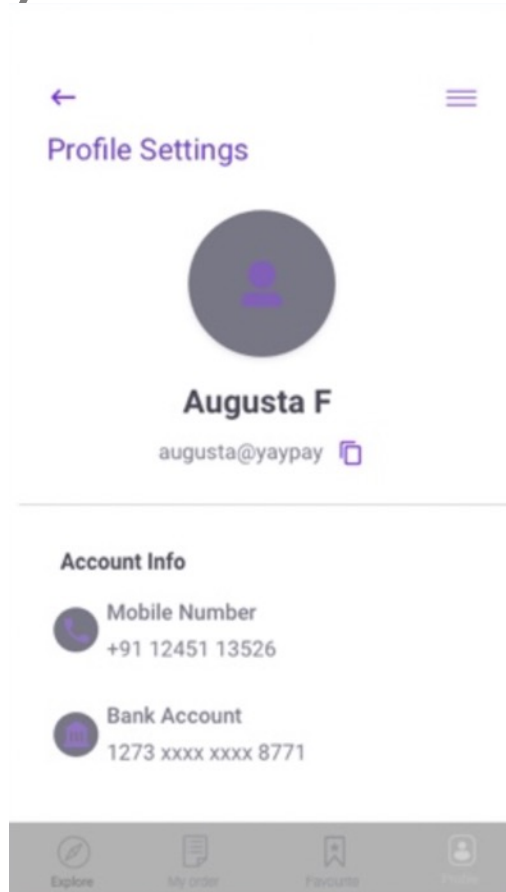
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[read more >](#)

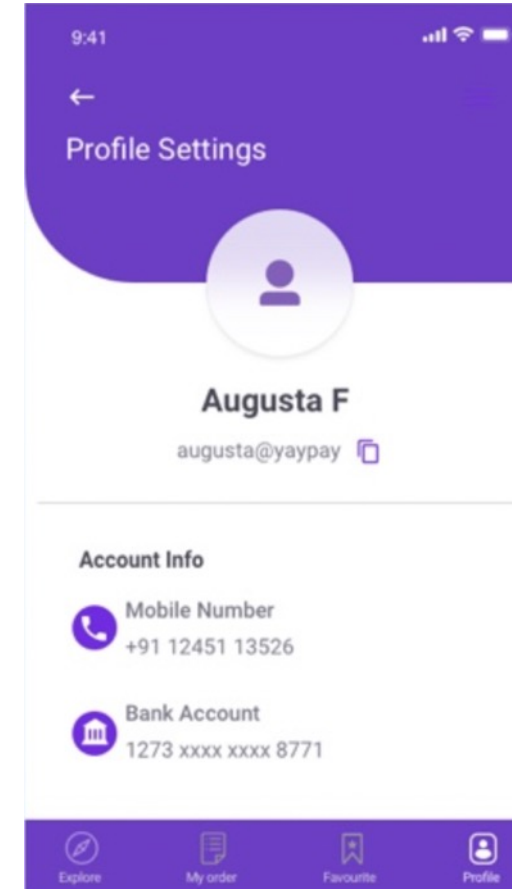
Option 2

Web Design

Question 7



Option 1



Option 2

Web Design

Question 8

Delicious meals for the whole family

These meals are guaranteed crowd-pleasers that even your children are bound to love, and all without breaking the bank.

Macaroni cheese

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Chicken pot pie

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nam

Option 1

Delicious meals for the whole family

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Donec finibus ex convallis urna placerat aliquet. Cras bibendum sem mauris, sollicitudin ultrices odio cursus sed. Pellentesque ac ex dignissim nulla gravida facilisis.

Chicken pot pie

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Option 2

Web Design

User Experience (UX)

- Focuses on the overall experience a user has with a website or product.
- It answers:
 - Is it easy to use?
 - Is the navigation intuitive?
 - Does it solve the user's problem?

Web Design

User Interface (UI)

- Focuses on the look and feel of the product.
- Includes:
 - Layout
 - Buttons
 - Icons
 - Colors
 - Typography
 - Spacing

Web Design

UX vs UI

UX is how it works.

UI is how it looks.

Web Design

Color Theory

Why Color Matters:

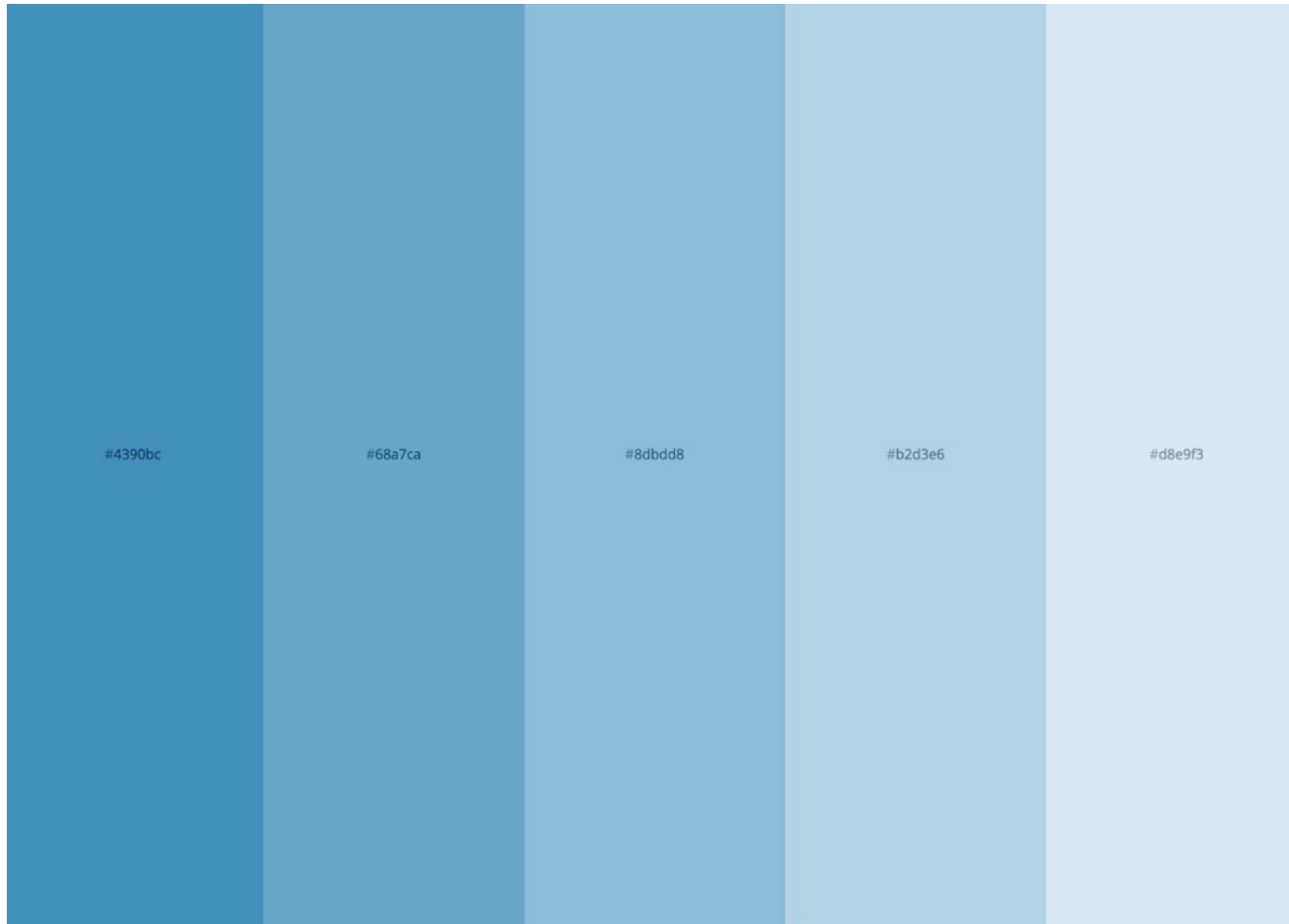
- Influences mood, attention, and user behavior.
- Builds brand identity and visual hierarchy.

Tips:

- Stick to a limited palette (2–4 main colors).
- Use contrast for readability (e.g., dark text on light background).
- Use accent colors for buttons, links, and CTAs.

Web Design

Color Theory - Monochrome



Web Design

Color Theory - Complementary



Web Design

Color Theory - Analogous



Web Design

Typography

Typography = Visual Language

Helps users read, understand, and feel the message.

Key Elements:

- Font Family: Use web-safe or Google Fonts (e.g., Roboto, Open Sans).
- Font Size: Prioritize readability; use consistent heading hierarchy (H1 > H2 > P).
- Use max 2 fonts per site (e.g., one for headings, one for body).
- Keep body text between 16–18px.
- Avoid decorative fonts for long text blocks.

Web Design

Principles of Good Web Design

1. Clarity & Simplicity

- Users should instantly know what to do.
- Avoid clutter and unnecessary elements.

2. Consistency

- Use consistent colors, fonts, and layouts across pages.

3. Responsiveness

- Design for **all devices** (desktop, tablet, mobile).

Web Design

Principles of Good Web Design

4. Visual Hierarchy

- Guide the eye using size, color, spacing, and alignment.

5. Accessibility

- Use alt text, color contrast, keyboard navigation, etc.

6. Feedback

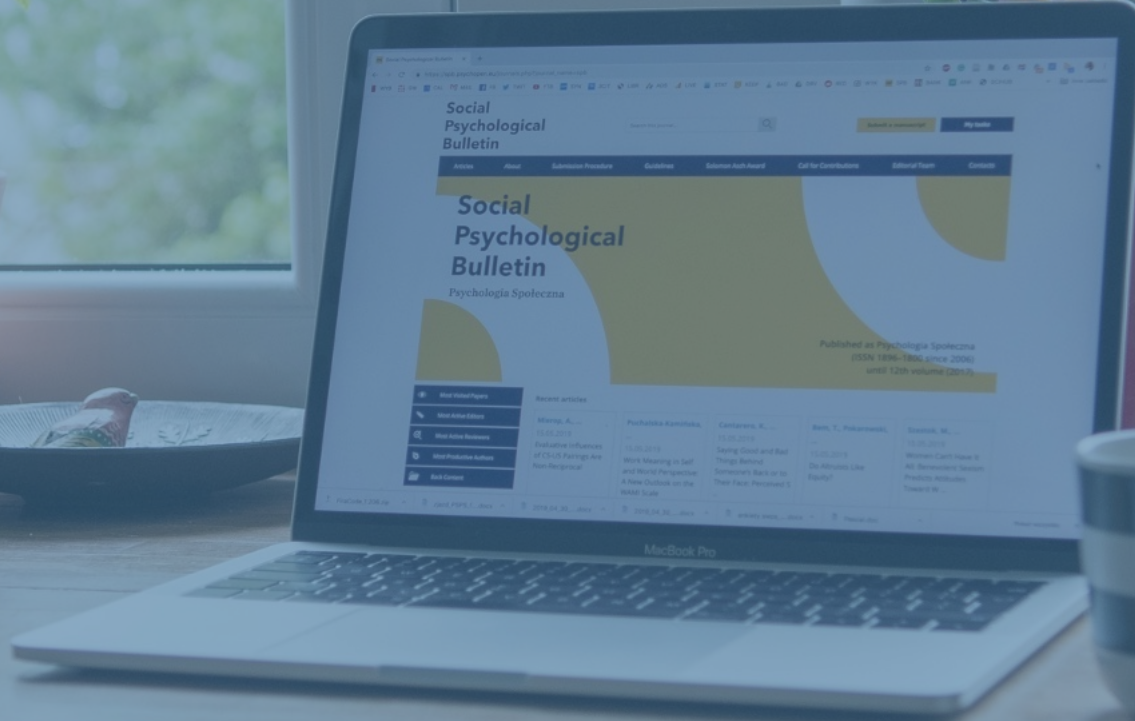
- Buttons and actions should give visual or audio feedback.

7. Speed

- Optimize images and assets; fast sites retain users.



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Any questions?